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Semiconductor Foundry Market (2026 - 2035)

Size, Share & Trends Analysis Report By Technology Node (10/7/5 nm & less, 12/11 nm, 16/14 nm, 20 nm), By Wafer Size, By Application, By Region, And Segment Forecasts,

Report Summary | Table of Contents | Segmentation | Methodology

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Market Size, 2025	Market Estimate, 2026	Market Forecast, 2035
$180.7B	$194.8B	$383.0B
CAGR, 2026–2035		

Report Details

- ID: GVR454884
- Report Pages: 100
- Format: PDF, Excel, Dashboard
- Published: August 2026

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- Market Overview
- Technology Node Insights
- Wafer Size Insights
- Application Insights

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7.8%

Semiconductor Foundry Market Summary

The global semiconductor foundry market size was valued at USD 180.7 billion in 2025 and is projected to grow from USD 194.8 billion in 2026 to USD 383.0 billion in 2035, at a CAGR of 7.8% from 2026 to 2035. Asia Pacific dominated the market, accounting for approximately 42.0% of revenue in 2025. The semiconductor foundry market is growing due to rising demand for advanced chips, AI applications, demand for advanced automotive electronics, and increasing investments in semiconductor manufacturing capacity.

Key Market Trends & Insights

- By technology node: The 10/7/5 nm and less segment held the largest market share of approximately 68.0% in 2025.
- By wafer size: The 300 mm segment held the largest market share of approximately 75.0% in 2025.
- By application: The consumer electronics and communication segment held the largest market share of approximately 32.0% in 2025.

Regional Highlights

- Largest regional market: Asia Pacific (42.0% revenue share, 2025)
- Fastest-growing regional market: North America (highest CAGR, 2026-2035)

Market Size & Forecast

- Market size in 2025: USD 180.7 Billion
- Estimated market size in 2026: USD 194.8 Billion

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- Projected market size by 2035: USD 383.0 Billion
- CAGR (2026-2035): 7.8%

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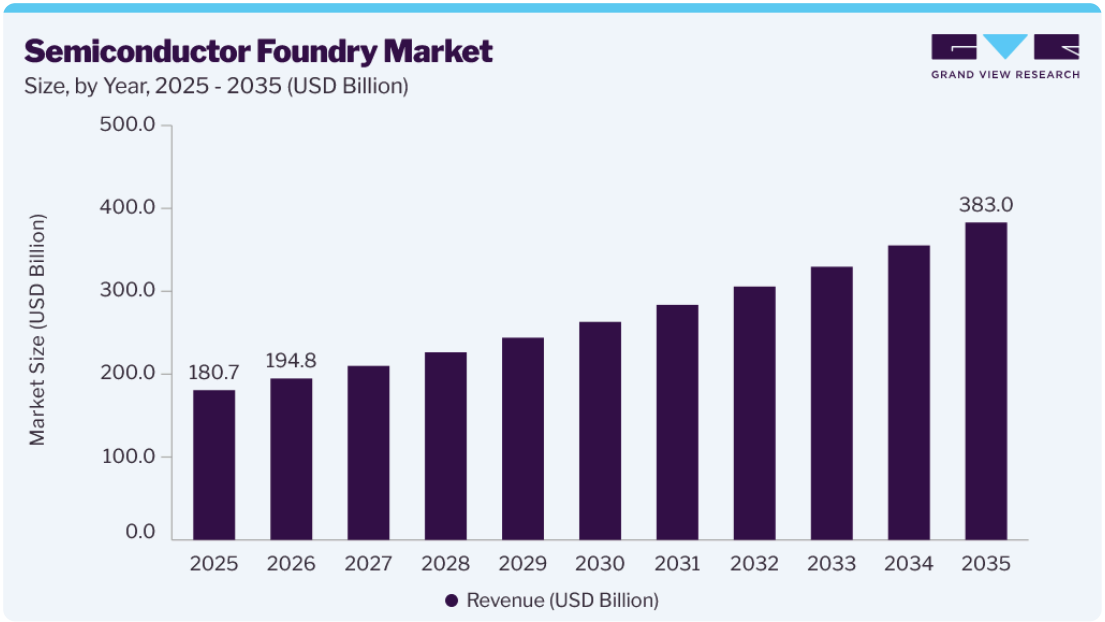
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The semiconductor foundry market is experiencing significant growth due to rising demand for advanced semiconductor chips across artificial intelligence, high-performance computing, data centers, consumer electronics, and automotive applications. The increasing adoption of smaller technology nodes, growing semiconductor content in electric vehicles, and expansion of connected devices are encouraging foundries to invest in advanced manufacturing capabilities. Additionally, the shift toward outsourced semiconductor manufacturing by fabless companies, increasing demand for high-performance processors, and government initiatives to strengthen domestic semiconductor production are creating new growth opportunities for the market.



Source: Grand View Research, IR Documents, Primary Interviews, Paid Databases

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However, the market faces challenges such as high capital investment requirements for establishing and upgrading fabrication facilities, increasing complexity of advanced node manufacturing, and supply chain disruptions affecting semiconductor production. The shortage of skilled workforce and dependence on specialized equipment and raw materials may also limit market expansion. Despite these restraints, opportunities arising from AI-driven chip demand, semiconductor localization strategies, expansion of automotive electronics, and increasing investments in next-generation fabrication technologies are expected to support long-term market growth.

Analyst Perspective

The semiconductor foundry market is expected to witness strong growth, driven by increasing demand for advanced semiconductor technologies across AI, high-performance computing, automotive electronics, and communication infrastructure. The transition toward smaller technology nodes and increasing investments in fabrication capacity are expected to remain key growth factors. Foundries are focusing on improving production efficiency, expanding advanced manufacturing capabilities, and strengthening supply chain resilience to address rising semiconductor demand. However, high capital expenditure requirements, technological complexity, and geopolitical uncertainties may create challenges for market participants. The growing adoption of electric vehicles, AI-enabled devices, and government-supported semiconductor manufacturing initiatives is expected to create significant opportunities. Companies with advanced process capabilities, scalable production capacity, and strong ecosystem partnerships are likely to gain a competitive advantage in the evolving semiconductor foundry landscape.

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Technology Node Insights

The 10/7/5 nm and below segment held the largest market share of approximately 68.0% and is also expected to register the fastest growth during the forecast period, driven by increasing demand for advanced semiconductor chips used in artificial intelligence, high-performance computing, data centers, and premium electronic devices. The continuous expansion of advanced process technologies is enabling higher performance, improved power efficiency, and greater transistor density, making smaller nodes essential for next-generation chip designs. Foundries are significantly increasing investments in leading-edge fabrication capacity to support AI-driven semiconductor demand. For instance, in July 2026, TSMC accelerated expansion of its 2 nm production capacity, with multiple fabs in Taiwan being ramped to address strong demand from AI and high-performance computing applications. The company's 2 nm family expansion, including future process enhancements, highlights the growing importance of advanced nodes in the semiconductor foundry industry.

The 16/14 nm segment is expected to witness steady growth due to its continued adoption in applications requiring a balance between performance, power efficiency, and manufacturing cost, including automotive electronics, industrial systems, communication devices, and mid-range consumer electronics.

Also, the 20 nm and more segment continues to support mature semiconductor applications such as microcontrollers, power management chips, and cost-sensitive electronic components. Although mature nodes are growing at a slower pace compared with advanced technologies, increasing demand for automotive, industrial, and Internet of Things (IoT) devices continues to sustain their relevance in the semiconductor foundry ecosystem.

Wafer Size Insights

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The 300 mm segment held the largest market share of approximately 75.0% and is expected to register the fastest growth during the forecast period, driven by its higher manufacturing efficiency, improved wafer yield, and cost advantages compared with smaller wafer sizes. Large-diameter wafers enable semiconductor foundries to produce a greater number of chips per wafer, reducing overall production costs and improving operational scalability. The growing demand for advanced semiconductor chips used in artificial intelligence, high-performance computing, data centers, automotive electronics, and consumer devices is accelerating the adoption of 300 mm wafer technology. Leading foundries are increasingly expanding 300 mm fabrication capacity to support high-volume production and advanced process technologies.

The 200 mm segment is expected to maintain steady growth due to continued demand for mature semiconductor technologies used in automotive, industrial, power management, and IoT applications. These wafers remain economically viable for specialized chips where advanced nodes are not required. The 150 mm and below segment continues to serve niche applications, including legacy semiconductor devices, sensors, and specialty chips. Although adoption is declining compared with larger wafer sizes, demand from specific industrial and emerging applications is expected to sustain the segment during the forecast period.

Application Insights

The consumer electronics and communication segment held the largest market share of approximately 32% in 2025, driven by widespread semiconductor adoption across smartphones, communication infrastructure, personal computing devices, and connected consumer electronics. The increasing demand for advanced processors, artificial intelligence-enabled devices, 5G connectivity, and high-performance chips is encouraging semiconductor foundries to expand

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manufacturing capacity and invest in advanced process technologies. Continuous digital transformation, increasing data consumption, and the growing penetration of connected devices are expected to sustain demand from this segment during the forecast period.

The automotive segment is expected to register the fastest growth during the forecast period, supported by rising semiconductor content in electric vehicles, autonomous driving systems, advanced driver assistance systems (ADAS), and connected vehicle platforms. The increasing complexity of automotive electronics is driving demand for advanced logic chips, sensors, microcontrollers, and power management semiconductors. The transition toward vehicle electrification, intelligent mobility, and software-defined vehicles is expected to create significant growth opportunities for semiconductor foundries. The industrial and high-performance computing and data centers segments are also expected to witness strong growth due to increasing adoption of automation, artificial intelligence workloads, cloud computing, and edge computing solutions.

Regional Insights

The Asia Pacific region held the largest market share of approximately 42.0 in 2025, supported by its strong semiconductor manufacturing ecosystem, presence of leading foundries, and well-established supply chain networks across Taiwan, South Korea, China, and Japan. The region has a significant advantage due to the availability of advanced fabrication facilities, experienced semiconductor workforce, and strong presence of semiconductor equipment and material suppliers. Increasing demand for consumer electronics, communication devices, artificial intelligence chips, and automotive semiconductors is driving continued investments in wafer fabrication capacity and advanced process technologies. The region's dominance is further supported by the presence of major semiconductor

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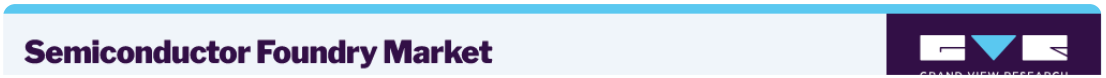
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manufacturing hubs and ongoing expansion of leading-edge and mature node production capabilities.



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North America Semiconductor Foundry Market Trends

The North America region is expected to register the fastest growth during the forecast period, driven by increasing focus on semiconductor supply chain diversification, government support for domestic chip manufacturing, and rising investments in fabrication infrastructure. The growing demand for AI processors, high-performance computing, data centers, and automotive semiconductors is encouraging foundries and technology companies to establish advanced manufacturing capabilities in the region. For instance, in March 2025, TSMC announced plans to expand its U.S. investment to USD 165 billion, including three additional fabrication plants, advanced packaging facilities, and an R&D center in

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Arizona, strengthening domestic semiconductor manufacturing capacity and supporting the region's growing role in the global foundry ecosystem. Increasing collaboration between semiconductor manufacturers, technology companies, and government initiatives is expected to further accelerate the growth of the North American semiconductor foundry market.

Key Semiconductor Foundry Company Insights

The semiconductor foundry market is characterized by intense competition, technological advancements, and continuous investments in advanced manufacturing capabilities. Key players are focusing on expanding fabrication capacity, developing smaller technology nodes, enhancing advanced packaging capabilities, and strengthening supply chain resilience to address rising demand from AI, high-performance computing, automotive, and consumer electronics applications. Companies are increasingly prioritizing strategic expansions, research and development initiatives, and partnerships to maintain technological leadership and support evolving semiconductor requirements. The competitive landscape is shaped by continuous innovation, manufacturing scalability, and the ability to deliver high-performance and cost-efficient semiconductor solutions across diverse applications.

Key Semiconductor Foundry Companies

The following key companies have been profiled for this study on the semiconductor foundry market:

- Taiwan Semiconductor Manufacturing Company Limited (TSMC)
- Samsung
- Intel Corporation

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- SMIC
- Hua Hong Grace Semiconductor Limited
- Tower Semiconductor
- Powerchip Semiconductor Manufacturing Corporation
- Vanguard International Semiconductor Corporation

Recent Developments

- In July 2026, Samsung Electronics and Broadcom expanded their strategic collaboration across memory and foundry technologies to accelerate the development of next-generation AI infrastructure. The collaboration focuses on combining Samsung's advanced semiconductor manufacturing capabilities with Broadcom's expertise in AI and networking solutions to support high-performance computing and AI-driven applications.
- In July 2026, The Government of India highlighted progress under the Semicon India Programme, with 12 semiconductor manufacturing projects approved across six states with total investment commitments of around ₹1.64 lakh crore. The initiative aims to strengthen domestic semiconductor manufacturing, packaging, and chip design capabilities while supporting India's development as a global electronics and semiconductor hub.

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- In April 2024, Wipro and Intel Foundry announced a collaboration to accelerate chip design and development by combining Wipro's semiconductor engineering capabilities with Intel Foundry's advanced manufacturing technologies. The partnership aims to support customers in developing customized semiconductor solutions and strengthen innovation across the global semiconductor ecosystem.

Semiconductor Foundry Market Report Scope

Report Attribute	Details
Market size in 2025	USD 180.7 billion
Estimated market size in 2026	USD 194.8 billion
Projected market size by 2035	USD 383.0 billion
Growth rate	CAGR of 7.8% from 2026 to 2035
Base year for estimation	2025
Actual data	2021 – 2024
Forecast period	2026 – 2035

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Quantitative units	Revenue in USD billion, and CAGR from 2026 to 2035
Report coverage	Revenue forecast, company ranking, competitive landscape, growth factors, and trends
Segments covered	Technology node, wafer size, application, region
Regional scope	North America; Europe; Asia Pacific; Latin America; Middle East & Africa
Country scope	U.S.; Canada; Mexico; UK; Germany; France; Italy; Spain; China; Japan; India; Australia; South Korea; Brazil; Argentina; South Africa; Saudi Arabia; UAE
Key companies profiled	Taiwan Semiconductor Manufacturing Company Limited (TSMC); Samsung; Intel Corporation; GlobalFoundries; United Microelectronics Corporation; SMIC; Hua Hong Grace Semiconductor Limited; Tower Semiconductor; Powerchip Semiconductor Manufacturing Corporation; Vanguard International Semiconductor Corporation

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Global Semiconductor Foundry Market Report Segmentation

This report forecasts revenue growth at the global, regional, and country levels and provides an analysis of the latest industry trends and opportunities across each sub-segment from 2021 to 2035. For this study, Grand View Research has segmented the global semiconductor foundry market report based on technology node, wafer size, application, and region:

- Technology Node Outlook (Revenue, USD Billion, 2021 - 2035)
  - 10/7/5 nm and less
  - 12/11 nm
  - 16/14 nm
  - 20 nm
- Wafer Size Outlook (Revenue, USD Billion, 2021 - 2035)

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- 300 mm
- 200 mm
- Less than 150 mm

- **Application Outlook (Revenue, USD Billion, 2021 - 2035)**
  - Consumer Electronics
  - Automotive
  - Industrial
  - IT & Telecom
  - Healthcare Devices
  - Others
- **Regional Outlook (Revenue, USD Billion, 2021 - 2035)**
  - North America
    - U.S.
    - Canada
    - Mexico
  - Europe

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This report was authored by the advanced interior materials research team at Grand View Research - comprising two research analysts, one senior research analyst, and one industry expert - with specialized expertise in the advanced interior materials segment of the advanced materials industry. All findings are based on proprietary advanced materials databases, executive interviews, and regulatory analysis, subject to internal peer review prior to publication.

Last Updated: September 2026

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Key questions answered by the report

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